

Group 1 Q&A - Answers

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Q&A

How much does night sky brightness vary on a daily (or even longer time period) basis? Are daily differences caused by any specific factor?

At a particular location and time of night, the night sky brightness typically does not vary that widely day-to-day. For large differences between two consecutive days, we hypothesize that the differences are most likely attributed to changes in the weather. See the following figures 1 & 2 showing the vaguely periodic behavior across the entire time epoch of the data, and then the more apparent periodic behavior for brightness by the hour relative to noon for a specific site (GMARS):

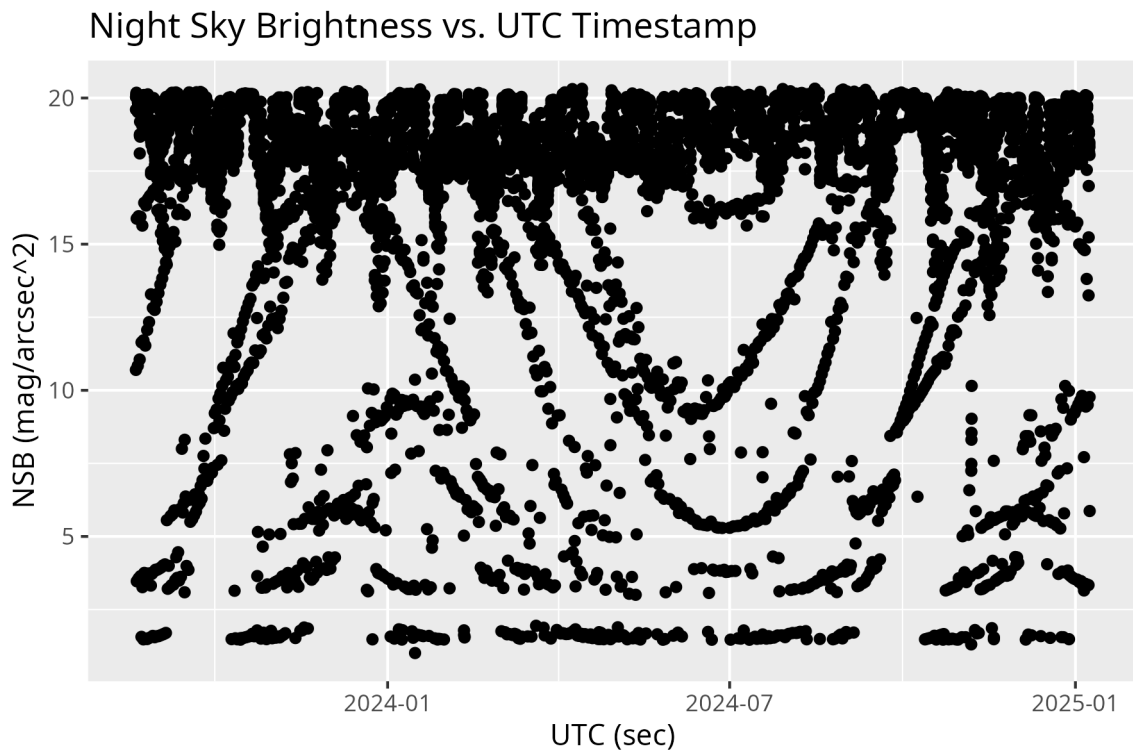


Figure 1: NSB v. UTC Time for all sites

Why do you think elevation was selected as an important feature?

Elevation is likely an important feature because the air is thinner at higher elevations, so there are less air particles for the light to bounce off of, likely leading to darker skies. See figure 3.

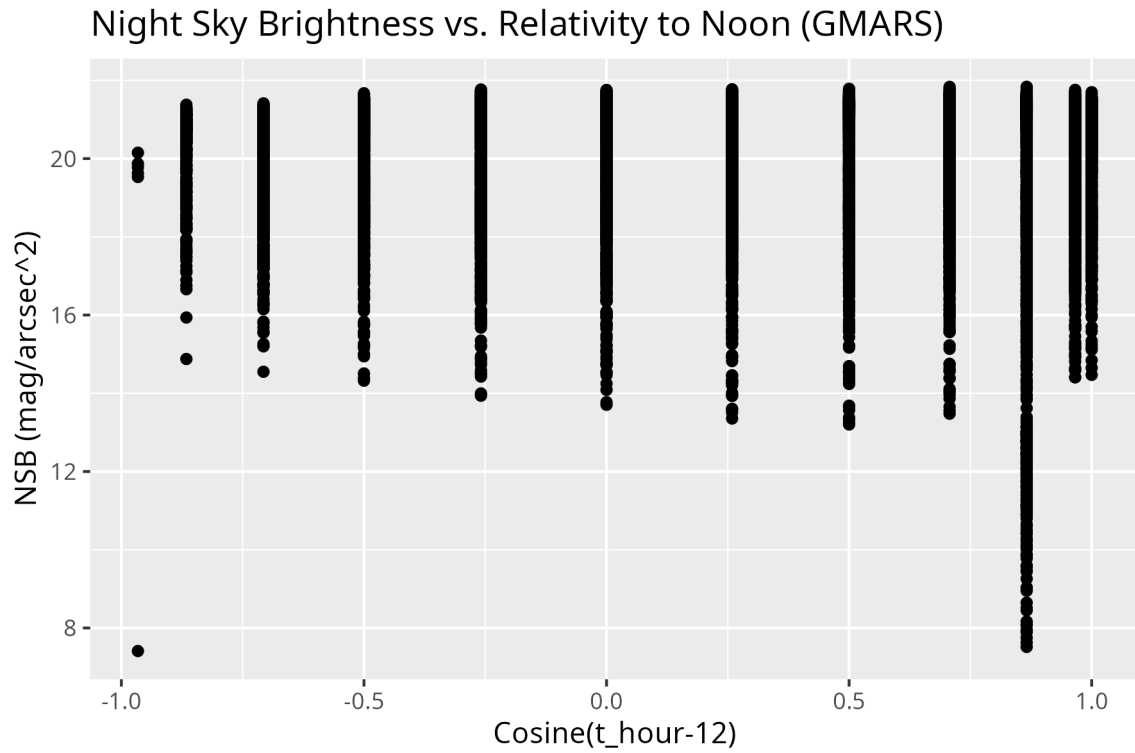


Figure 2: NSB v. Time relative to Noon for the GMARS site

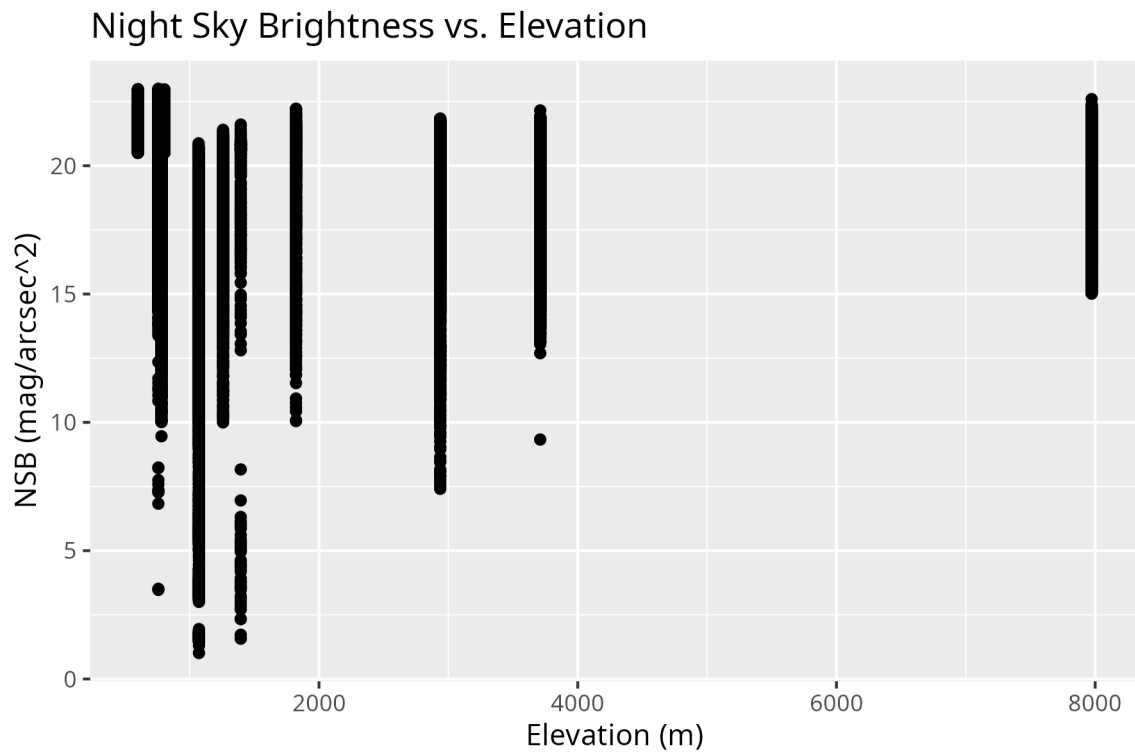


Figure 3: NSB v. Elevation

What is the importance of predicting light pollution other than wanting to see the stars? Would your models need to be adjusted for other applications?

Modeling light pollution could be useful for locating new potential observatory sites, or studying the potential future impact that light pollution could have on current observatories and astronomical science.